

Emotion-Aware Chatbot using Transformer and Sentiment Conditioning

Vijaysai Ananthavaram

Deep Learning Project

Professor John Glossner

May 06, 2026

Abstract

This project focuses on improving chatbot systems by making them aware of user emotions during conversations. Traditional chatbots usually respond only based on the text content without understanding how the user feels. Because of this, their responses often appear generic and lack empathy. The main goal of this project is to design a system that can first detect the emotional state of a user from their message and then generate a response that aligns with that emotion. The system uses transformer-based deep learning models for emotion classification and integrates this information into response generation. The approach follows a two-stage pipeline where emotion detection is performed first, followed by response generation guided by the detected emotion. The performance of the system is evaluated using both quantitative metrics such as accuracy, F1-score, and BLEU score, as well as qualitative analysis of generated responses. The results show that the emotion-aware chatbot produces more meaningful and contextually appropriate replies compared to a baseline chatbot. This work highlights the importance of emotional intelligence in conversational AI and demonstrates how deep learning techniques can enhance user interaction.

Table of Contents

1. Problem Statement and Motivation	4
2. Data or Environment Description	4
3. Method and Model Design	5
6. Failure Analysis and Limitations.....	8
7. Ethics, Limitations, and Responsible Use	9
8. Conclusion	10
9. References	10

1. Problem Statement and Motivation

Many modern chatbot systems are designed to process user queries and provide answers based on textual information. However, these systems often ignore the emotional context of the conversation. This creates a gap between human communication and machine-generated responses. For example, when a user expresses sadness or frustration, a typical chatbot may respond in a neutral tone, which can feel inappropriate or unhelpful. This limitation reduces user satisfaction and makes interactions less natural.

The problem addressed in this project is the lack of emotional awareness in chatbot systems. The objective is to build a system that can recognize emotions such as happiness, sadness, anger, fear, and others from user input and use this information to generate more suitable responses. This involves both classification and conditional response generation.

The motivation behind this work comes from real-world applications where understanding emotions is essential. In areas such as online education, customer service, and personal assistance, users expect systems to respond in a way that reflects their feelings. A chatbot that can recognize and respond to emotions can provide better support and improve the overall user experience.

Another important motivation is the advancement of deep learning models, especially transformers, which have shown strong performance in natural language processing tasks. These models provide an opportunity to build systems that can understand both semantic meaning and emotional context.

The success of this project is measured not only by accuracy in emotion detection but also by the quality of generated responses. The system should demonstrate improvement over a baseline model that does not use emotion information.

2. Data or Environment Description

The project uses publicly available datasets to train and evaluate the models. For emotion

classification, datasets containing text labeled with emotion categories are used. These datasets include multiple emotion classes such as joy, sadness, anger, fear, love, and surprise. The diversity of these labels helps the model learn different emotional patterns present in human language.

For conversational response generation and evaluation, datasets such as DailyDialog are used. These datasets contain structured conversations where each input message is paired with a suitable response. This helps the system learn how conversations flow and how responses should be constructed.

The dataset is divided into training, validation, and test sets using a standard split. The training set is used to learn model parameters, the validation set is used for tuning hyperparameters, and the test set is used for final evaluation. This ensures that the results reflect real-world performance and are not biased by training data.

Preprocessing is an important step in preparing the data. The text is cleaned by removing unnecessary spaces and normalizing formatting. Tokenization is performed to convert text into numerical representations that can be processed by transformer models. Padding and truncation are applied to ensure consistent input length.

The implementation environment includes Python and deep learning frameworks such as PyTorch and HuggingFace Transformers. These tools provide efficient and scalable methods for training and deploying transformer models. The system is designed to run on both CPU and GPU environments, making it accessible for different hardware setups.

3. Method and Model Design

The system is designed as a two-stage pipeline that combines emotion detection with response generation. This design allows the system to first understand the emotional context and then produce a response that reflects that context.

In the first stage, a transformer-based model such as DistilBERT is used for emotion

classification. This model takes a user's message as input and outputs a probability distribution over emotion classes. The predicted emotion is selected based on the highest probability. Transformer models are used because they can capture relationships between words and understand context effectively.

In the second stage, the detected emotion is used to guide the response generation process.

The chatbot combines the original user input with the predicted emotion to generate a response. This is known as sentiment conditioning, where additional information about emotion influences the output. This approach ensures that the generated response is not only relevant to the content but also aligned with the user's emotional state.

A baseline chatbot is implemented for comparison. This chatbot generates responses using simple rules without considering emotions. By comparing this baseline with the emotion-aware chatbot, the impact of emotion integration can be clearly observed.

The overall model design focuses on simplicity and effectiveness. Instead of building a highly complex system, the project demonstrates how combining existing deep learning techniques can lead to meaningful improvements in conversational systems.

4. Experimental Setup

The experimental setup is designed to evaluate both the emotion classification model and the chatbot system. The first step involves training the emotion classifier using labeled data. The training process uses standard optimization techniques such as the Adam optimizer and cross-entropy loss. The model is evaluated after each epoch to monitor performance.

Hyperparameters such as learning rate, batch size, and number of epochs are tuned using the validation set. This helps in achieving a balance between model performance and training efficiency. The system is configured to run efficiently even on limited hardware.

After training the classifier, the chatbot system is built by integrating the emotion detection module with response generation. Both the baseline chatbot and the emotion-aware chatbot

are tested using the same set of prompts. This ensures that the comparison is fair and consistent.

Evaluation is performed using both automatic and manual methods. Automatic evaluation includes accuracy and F1-score for emotion detection and BLEU score for response generation. Manual evaluation involves reviewing responses to check whether they are meaningful and aligned with the detected emotion.

The experiments are repeated to ensure stability of results. This helps in confirming that the improvements observed are consistent and not due to random variation.

5. Results

The results show that the emotion-aware chatbot performs better than the baseline chatbot in generating appropriate responses. The emotion classification model achieves good accuracy and F1-score, indicating that it can reliably identify emotions in user input.

When comparing responses, the baseline chatbot tends to produce generic replies that do not reflect the emotional context. In contrast, the emotion-aware chatbot generates responses that are more aligned with the user's feelings. For example, when a user expresses sadness, the emotion-aware chatbot provides a supportive response, while the baseline chatbot gives a neutral reply.

The BLEU score results indicate that the emotion-aware model produces responses closer to expected outputs. However, BLEU scores alone do not capture the full quality of conversational responses, so manual evaluation is also considered.

Manual analysis shows that the emotion-aware chatbot provides more natural and meaningful interactions. Users are more likely to feel understood when the chatbot responds with emotional awareness.

Overall, the results confirm that incorporating emotion into chatbot systems improves both response quality and user experience.

	model	model_source	accuracy	precision_weighted	recall_weighted	f1_weighted	f1_macro	matched_labeled_prompts	labeled_prompts	elapsed_seconds	rank
2	hartmann	j-hartmann/emotion-english-distilroberta-base	1.0000	1.0000	1.0000	1.0000	1.0000	3	3	0.17	1
3	local	saved_emotion_model	0.6667	0.5000	0.6667	0.5555	0.5555	3	3	0.18	2

Figure 1: Comparison of model performance using accuracy, precision, recall, and F1-score across different chatbot configurations

Search this file													
	model	model_source	status	matched_labeled_prompts	labeled_prompts	match_rate	accuracy	precision_weighted	recall_weighted	f1_weighted	f1_macro	elapsed_seconds	err
2	local	saved_emotion_model	ok	2	3	0.6667	0.6667	0.5000	0.6667	0.5555	0.5555	0.18	
3	hartmann	j-hartmann/emotion-english-distilroberta-base	ok	3	3	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.17	

Figure 2: Summary of model performance based on correct emotion predictions across evaluation prompts

	model	model_source	id	text	expected_emotion	predicted_emotion	raw_label	emotion_score	emotion_match
2	local	saved_emotion_model	sad_1	I feel really down today and I do not know how to get motivated.	sadness	sadness	sadness	0.3525	True
3	local	saved_emotion_model	joy_1	I just found out I got the internship I wanted.	joy	joy	joy	0.6342	True
4	local	saved_emotion_model	anger_1	I am upset because my teammate ignored all my messages.	anger	sadness	sadness	0.4054	False
5	hartmann	j-hartmann/emotion-english-distilroberta-base	sad_1	I feel really down today and I do not know how to get motivated.	sadness	sadness	sadness	0.9797	True
6	hartmann	j-hartmann/emotion-english-distilroberta-base	joy_1	I just found out I got the internship I wanted.	joy	joy	joy	0.6337	True
7	hartmann	j-hartmann/emotion-english-distilroberta-base	anger_1	I am upset because my teammate ignored all my messages.	anger	anger	anger	0.9675	True

expected_emotion	predicted_emotion	raw_label	emotion_score	emotion_match	baseline_response	emotion_aware_response
sadness	sadness	sadness	0.3525	True	I understand. Tell me more about that.	I'm sorry you're dealing with that. Do you want to share what has been hardest?
joy	joy	joy	0.6342	True	Hello. What would you like to talk about today?	That sounds really positive. What made you feel that way?
anger	sadness	sadness	0.4054	False	I understand. Tell me more about that.	I'm sorry you're dealing with that. Do you want to share what has been hardest?
sadness	sadness	sadness	0.9797	True	I understand. Tell me more about that.	I'm sorry you're dealing with that. Do you want to share what has been hardest?
joy	joy	joy	0.6337	True	Hello. What would you like to talk about today?	That sounds really positive. What made you feel that way?
anger	anger	anger	0.9675	True	I understand. Tell me more about that.	I can tell this is frustrating. Do you want to talk through what happened?

Figure 3: Sample comparison between baseline and emotion-aware chatbot responses for selected inputs

6. Failure Analysis and Limitations

Despite the improvements, the system has several limitations. One major limitation is that the quality of responses depends heavily on the accuracy of emotion detection. If the emotion is predicted incorrectly, the chatbot may generate an inappropriate response.

Another limitation is that emotions are complex and cannot always be clearly categorized.

Some user inputs may contain mixed emotions or subtle expressions that are difficult to detect. This can lead to incorrect predictions and affect response quality.

The dataset used for training may also introduce limitations. If the dataset is not diverse enough, the model may not generalize well to different types of users or situations. This can result in biased or less effective responses.

The system also relies on predefined response patterns, which may limit its ability to generate highly creative or context-specific replies. In some cases, the responses may appear repetitive or less detailed.

These limitations highlight the need for further research and improvement in both emotion detection and response generation.

7. Ethics, Limitations, and Responsible Use

This project involves analyzing user emotions, which raises important ethical considerations.

One major concern is the potential misuse of the system in sensitive situations. The chatbot should not be used for applications such as medical advice or mental health support, where incorrect responses could have serious consequences.

Another important issue is bias in the dataset. If the training data does not represent a wide range of users and situations, the model may produce biased responses. This can affect fairness and reliability.

Privacy is also a key concern. User inputs may contain personal or sensitive information, so proper data handling practices must be followed to ensure confidentiality.

The system should be used as a support tool rather than a decision-making system. Users should be informed about its limitations and should not rely on it for critical decisions.

Responsible deployment of such systems requires transparency, fairness, and careful monitoring to prevent misuse.

8. Conclusion

This project demonstrates how deep learning can be used to improve chatbot systems by incorporating emotional understanding. By combining emotion detection and response generation, the system produces more meaningful and context-aware responses.

The comparison between the baseline and emotion-aware chatbot clearly shows the benefits of using emotion information. The results indicate that emotional awareness improves both the quality of responses and user satisfaction.

The project also highlights the challenges involved in building such systems, including limitations in data and model performance.

Overall, this work provides a strong foundation for future research in emotion-aware conversational systems. Further improvements can focus on better emotion detection, more advanced response generation, and handling complex emotional expressions.

9. References

Devlin, J., Chang, M. W., Lee, K., & Toutanova, K. (2019). BERT: Pre-training of deep bidirectional transformers for language understanding. Proceedings of NAACL-HLT.

Li, Y., Su, H., Shen, X., Li, W., Cao, Z., & Niu, S. (2017). DailyDialog: A manually labelled multi-turn dialogue dataset. Proceedings of IJCNLP.

Papineni, K., Roukos, S., Ward, T., & Zhu, W. J. (2002). BLEU: a method for automatic evaluation of machine translation. Proceedings of ACL.

Wolf, T., Debut, L., Sanh, V., et al. (2020). Transformers: State-of-the-art natural language processing. Proceedings of EMNLP: System Demonstrations.